AI for Medicine – Project Report

Prof. Stefano Diciotti – University of Bologna

Academic Year: 2025/2026

Student Name: Bonafè Carolina, Girotti Lorenzo

Degree Program: Artificial Intelligence

Submission Date: 19/08/2026

Note: The report should be concise but complete. Typical length: 8–10 pages.

1. Project Title

Diabetic Retinopathy Severity Grading: A Comparative Study of Lightweight Architectures and Ordinal Regression

2. Problem Statement

Diabetic Retinopathy (DR) is a leading cause of preventable blindness worldwide, affecting over 100 million people (Congdon, N. et al., 2012). The disease progresses through a well-defined ordinal severity scale (Grade 0 to 4), and early detection through retinal fundus screening is critical for timely intervention (Lee, R. et al., 2015).
In order to help the scaling of screening, automated DR grading systems must balance diagnostic accuracy with computational efficiency for practical deployment. At the same time, the ordinal nature of the severity scale, where misclassifying a Grade 4 as Grade 0 is far more harmful than confusing adjacent grades, may require training objectives that explicitly account for this structure. This project addresses these dual challenges of computational constraints and ordinal classification.

3. Objective of the Study

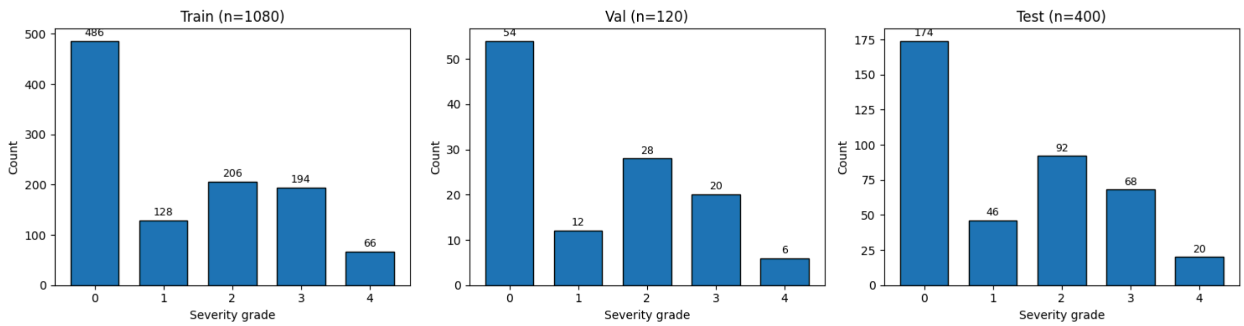
The primary objective of this project is to predict the 5-grade severity of DR from retinal fundus images by analyzing how model capacity and ordinal constraints interact in a severely imbalanced context. To this end, we compare two distinct architectures: a lightweight custom CNN (under 2M parameters) and a pre-trained EfficientNet-V2-RW-T (Tan, M. et al., 2021).
Both models are evaluated using standard Cross-Entropy and CORAL ordinal loss (Cao, W. et al., 2019), an approach that penalizes predictions proportionally to their distance from the true grade. Furthermore, we assess real-world clinical utility by reframing the task as binary screening with sensitivity-maximizing thresholds, and interpret model decisions through gradient-based explainability using HiResCAM (Draelos, R. L. et al., 2021).

4. Dataset Description

This project employs RetinaMNIST, a dataset from the MedMNIST v2 collection (Yang, J. et al., 2022). RetinaMNIST is derived from the DeepDRiD challenge (Liu R. et al., 2022) and consists of 1,600 retinal fundus images for 5-level diabetic retinopathy severity grading.

It is built upon RGB fundus images, originally acquired at 1,736 × 1,824 pixels, center-cropped to the short edge length and resized to 224 × 224 pixels. Labels follow an ordinal 5-class classification:

0. No DR: no apparent retinopathy
1. Mild NPDR: microaneurysms only
2. Moderate NPDR: more than microaneurysms but less than severe NPDR
3. Severe NPDR: extensive hemorrhages or venous beading, no PDR signs
4. Proliferative DR: neovascularization or vitreous hemorrhage



The dataset exhibits severe class imbalance, with Grade 0 (healthy) representing approximately 45% of the training samples, while the most severe class (Grade 4) accounts for only 6%. However, this imbalance is clinically realistic, where healthy and moderate cases are inherently more prevalent in screening populations (Ting, D. S. W. et al., 2017), but poses a challenge for model training, as the classifier may bias toward the majority class.

5. Data Preprocessing

Before training, each retinal fundus image undergoes a two-step preprocessing pipeline:

- **Ben Graham enhancement.** Following common practice in DR grading (Babu, S. et al., 2024; Ramalakshmi, D. et al., 2025), we apply an unsharp masking filter to enhance lesion contrast while preserving retinal structure. The filter computes the difference between the original image and its Gaussian-blurred version ($\sigma = 5$ according to the input image resolution) and amplifies the high-frequency components via:
  $I_{\text{processed}} = 4 \cdot I_{\text{original}} - 4 \cdot G_\sigma(I_{\text{original}}) + 128$
  This step is followed by a circular mask centered on the image that zeros out background pixels outside the fundus region, reducing noise from non-retinal areas. The resulting images are clipped to the $[0, 255]$ and then saved to disk as PNG images.
- **Normalization.** Per-channel mean and standard deviation are computed from the preprocessed training set and used to normalize all splits (train, validation, test) to zero mean and unit variance.

6. Avoiding Data Leakage

The entire pipeline is designed to prevent data leakage at every stage:

- **Test set isolation.** The test set is used exclusively for final evaluation. All hyperparameters and design choices were tuned empirically using the validation set only, without exposing the test labels during model development.
- **Cross-validation integrity.** During 5-fold cross-validation, class weights (for Cross-Entropy) and CORAL importance weights are computed strictly on the training portion of each fold, ensuring the held-out validation split remains completely unseen to the model until evaluation.
- **Fold-specific normalization.** Per-channel mean and standard deviation for image normalization are recomputed independently within each fold using only the training images of that fold, in order to avoid information leakage from the current validation set.
- **Duplicate detection.** An MD5 hash check was performed across all splits to verify that no identical images appear in multiple splits. No duplicates were found, which is expected given the small and well-maintained nature of the dataset.
- **Threshold tuning.** The operating point selection for the binary screening task (referable vs. non-referable) is also performed on the validation set, keeping the test set untouched for an unbiased final evaluation.

7. Machine Learning Pipeline

The machine learning pipeline is designed to ensure rigorous evaluation, prevent data leakage, and handle the severe class imbalance inherent to this DR grading task.

7.1. Models Used

The pipeline evaluates and compares two distinct neural network architectures:

- **Custom CNN (MicroSENet).** A lightweight convolutional neural network (~1.7M parameters) specifically designed for this task. It integrates *Squeeze-and-Excitation (SE)* blocks (Hu, J. et al., 2018) for channel-wise feature weighting, cross-stage feature fusion (combining outputs from Block 2 and Block 5) to capture both fine-grained lesions and deep semantic context, and a feature compression bottleneck that projects the representation into a 256-dimensional space before the final classification head, reducing model complexity and improving regularization.
- **EfficientNetV2-RW-T.** A custom tiny (13.6M parameter) variant of the EfficientNetV2 image classification model (Tan & Le, 2021) implemented within the PyTorch Image Models (`timm`) library. The EfficientNetV2-RW-T backbone sequentially processes inputs starting from an initial stem convolution, followed by a sophisticated hierarchy of *Edge Residual* and *Inverted Residual* mobile blocks.

  For this architecture, the framework incorporates modular layer freezing protocols to govern how the network adapts to target datasets under different training strategies. Under *linear probing*, all backbone parameters are frozen with gradient updates disabled, restricting training exclusively to the final classification head. In contrast, *partial fine-tuning* selectively unlocks the deepest convolutional block alongside the classification head, enabling task-specific refinement of high-level semantic representations.

Both architectures are evaluated using two different objective functions: Cross-Entropy (CE) for standard multi-class classification, and CORAL loss to explicitly enforce ordinal regression constraints on the severity grades.

7.3. Validation Techniques

A *Stratified K-Fold Cross-Validation* scheme, *5 folds* with shuffling enabled, is employed to evaluate model generalization while preventing data leakage. Stratification preserves the original class distribution of DR severity grades (0 to 4) across both training and validation splits for every fold, preventing unrepresentative splits in rare classes. For each fold, the model is initialized from scratch to prevent information transfer, and final performance is reported as mean ± standard deviation across all folds.

7.4. Evaluation Metrics

Accuracy, macro-averaged ROC-AUC, and Quadratic Weighted Kappa (QWK) are used to evaluate model performance. The macro-averaged ROC-AUC, computed using a One-vs-Rest (OvR) strategy, treats all classes equally (same weight) despite imbalance, while QWK penalizes predictions proportionally to their distance from the true grade, making it particularly informative for ordinal tasks. A normalized confusion matrix and classification report are additionally reported for each validation and test evaluation.

For the binary screening task, four candidate operating points are considered: a standard default threshold of 0.5 (where a summed predicted probability greater than 0.5 for grades 2-4 classifies a case as referable, which conversely corresponds to a probability of $\leq$ 0.5 for grades 0-1), the Youden Index (Youden, W. J., 1950), a sensitivity-maximizing threshold (Se $\geq$ 0.95), and a specificity-maximizing threshold (Sp $\geq$ 0.95). The final evaluation is based on the Se $\geq$ 0.95 threshold, prioritizing patient safety by minimizing false negatives. Performance is reported in terms of sensitivity, specificity, PPV, and NPV.

7.5. Regularization and Hyperparameter Tuning

Optimization and scheduling. All models across both the custom CNN and the EfficientNetV2-RW-T architectures are trained using the AdamW optimizer paired with a Cosine Annealing scheduler, which decays the learning rate to a minimum of 10^{-6} over a maximum of 50 epochs with a batch size of 32. Training is regularized with early stopping based on validation AUC using a patience of 10 epochs.

For the custom CNN, optimization relies on a standard AdamW configuration with a base learning rate of 3×10^{-4} and a weight decay of 10^{-4}. In contrast, the EfficientNetV2-RW-T models dynamically adapt their optimization strategy depending on the selected fine-tuning regime. Under linear probing, updates are restricted exclusively to the classifier parameters using an AdamW setup with a learning rate of 3×10^{-4} and weight decay of 10^{-4}. Under partial fine-tuning, the network employs differential learning rates and stronger regularization through parameter groups: unfrozen backbone layers receive a conservative learning rate of 2×10^{-5}, while the classifier head is updated at 10^{-4}, both paired with the same weight decay of 10^{-4}.

Dynamic imbalance weighting. To address class imbalance, fold-specific weights are computed depending on the loss function. For CE loss, class weights are computed using balanced inverse class frequency:

$$w_c = \frac{N}{K \cdot N_c}$$

where N is the total number of training samples in the fold, $K = 5$ is the number of classes, and N_c is the number of samples for class c. For CORAL loss, task-specific importance weights (Cao et al., 2019, Eq. 7) balance the uneven ratio of positive to negative samples across each of the 4 sequential binary thresholds.

Structural regularization. The custom CNN employs spatial dropout ($p = 0.1$) within the convolutional backbone and a stronger dropout ($p = 0.5$) in the classification head. The SE reduction ratio of 16 follows the original SE-Net (Hu, J. et al., 2018). In a similar way, also the EfficientNetV2-RW-T architecture incorporates a dropout layer of $p = 0.5$ placed between the backbone and the classification head across all training regimes and loss configurations.

8. Results

8.1. Cross-Validation Results

Model	Accuracy	AUC	QWK
MicroSENet (CE)	0.5093 ± 0.0356	0.8209 ± 0.0115	0.6727 ± 0.0265
MicroSENet (CORAL)	0.5102 ± 0.0269	0.7475 ± 0.0160	0.6452 ± 0.0280
EfficientNetV2 (CE, partial fine-tuning)	**0.5685 ± 0.0333**	**0.8549 ± 0.0123**	**0.7225 ± 0.0238**
EfficientNetV2 (CORAL, partial fine-tuning)	0.5019 ± 0.0063	0.7623 ± 0.0136	0.6631 ± 0.0209

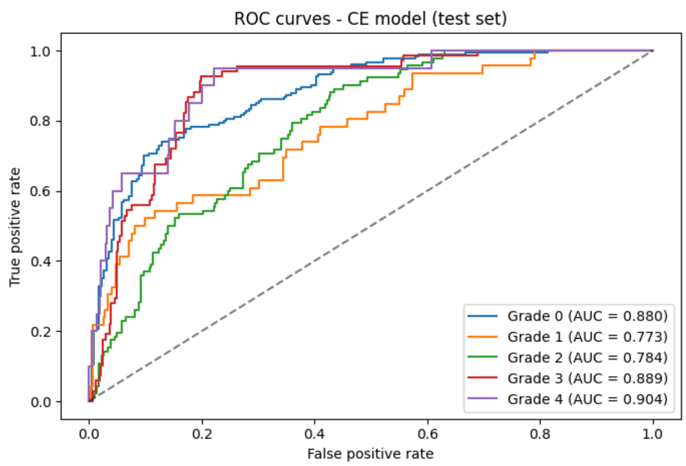
For EfficientNetV2, following preliminary validation and cross-validation where partial fine-tuning significantly outperformed linear probing, all subsequent experiments relied exclusively on this strategy .

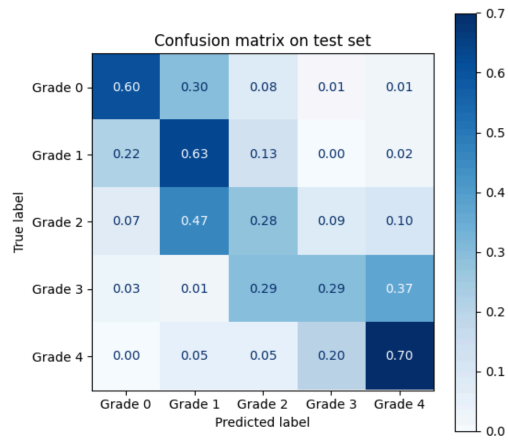
8.3. Test Set Results

Model	Accuracy	AUC	QWK	Macro F1-score
MicroSENet (CE)	0.4850	0.8459	**0.7376**	0.4323
MicroSENet (CORAL)	0.5125	0.7769	0.7110	0.3427
EfficientNetV2 (CE, partial fine-tuning)	**0.5450**	**0.8551**	0.7076	**0.4777**
EfficientNetV2 (CORAL, partial fine-tuning)	0.4850	0.7503	0.6452	0.2939

8.4. ROC Curves and Confusion Matrices (test set)

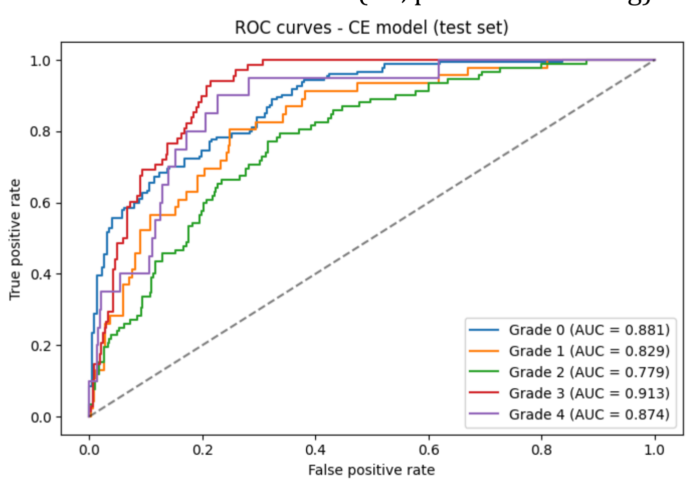
- MicroSENet (CE)

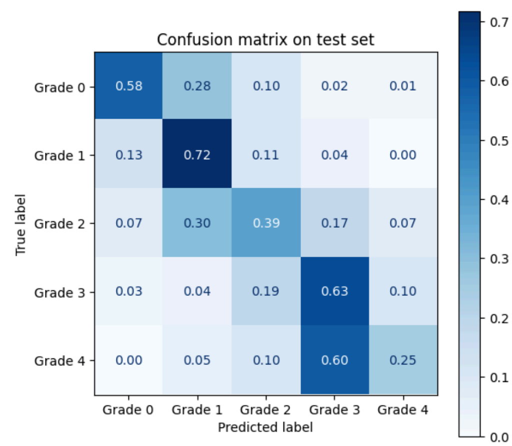




- MicroSENet (CORAL) omitted for document brevity, available in the notebook

- EfficientNetV2-RW-T (CE, partial fine-tuning)

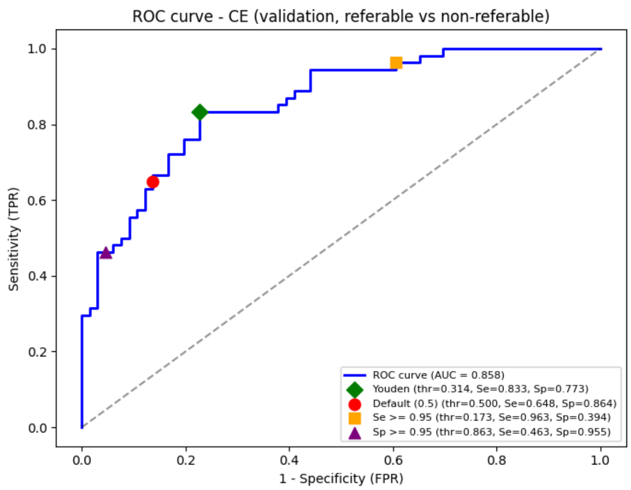


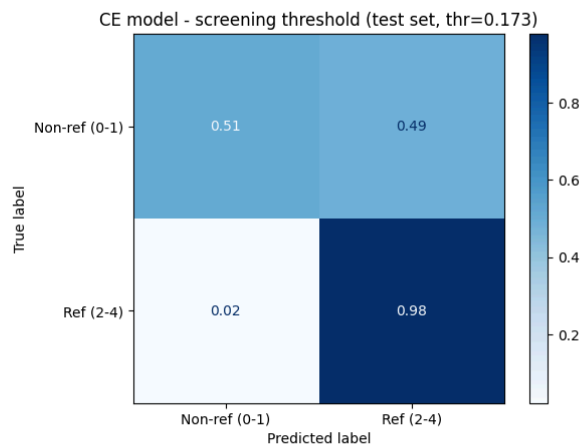


- EfficientNetV2-RW-T (CORAL, partial fine-tuning) omitted for document brevity, available in the notebook

8.5. ROC Curves and Confusion Matrices for the Referral Binary Task

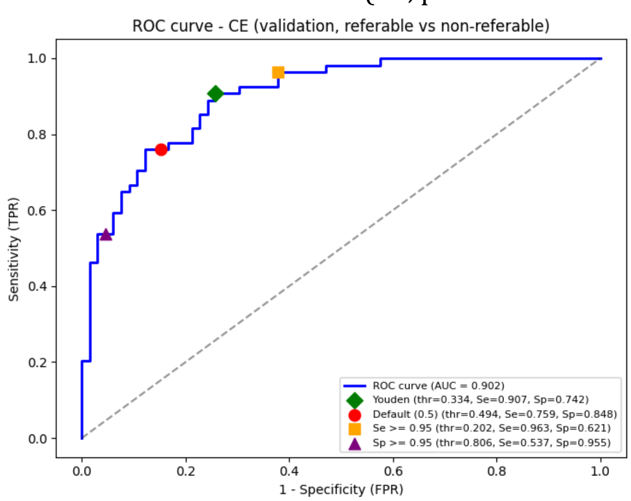
- MicroSENet (CE)

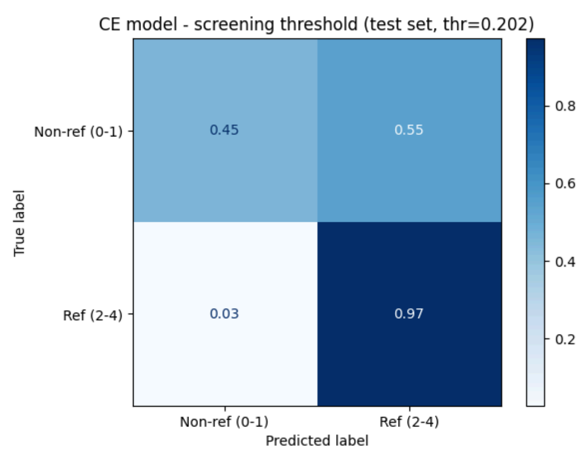




- MicroSENet (CORAL) omitted for document brevity, available in the notebook

- EfficientNetV2-RW-T (CE, partial fine-tuning)





- EfficientNetV2-RW-T (CORAL, partial fine-tuning) omitted for document brevity, available in the notebook

9. Discussion

Regarding MicroSENet, experimental results show a clear trade-off between the two training objectives. On the test set, the Cross-Entropy model achieves a higher AUC (0.8459), QWK (0.7376), and macro F1 (0.4323). The confusion matrix confirms this behavior, where CE maintains reasonable diagonal dominance across all classes, with misclassifications mostly confined to adjacent grades, showing particular strength in identifying early-stage (Grade 0 and 1) and severe (Grade 4) cases.

On the other hand, the CORAL model achieves a higher accuracy (0.5125) and a slightly lower QWK (0.7110) than with CE, not completely confirming the benefit of ordinal modeling. Indeed, the model often over-predicts Grade 0 images, nearly collapsing on Grade 1 which is almost entirely absorbed into the healthy class. This behavior can be explained by the ordinal regression formulation, because CORAL learns cumulative thresholds that are difficult to calibrate for intermediate classes when the majority class is concentrated at one extreme of the ordinal scale.

Regarding binary screening analysis, when the task is reframed as referable (Grades 2-4) versus non-referable (Grades 0-1) and a sensitivity-maximizing threshold ($Se \geq 0.95$) is applied, the CORAL model outperforms CE in specificity, PPV, and NPV, while matching its sensitivity: sensitivity (0.978 vs. 0.978), specificity (0.609 vs. 0.514), PPV (0.672 vs. 0.622), and NPV (0.971 vs. 0.966). This is a direct consequence of binarization, where CORAL's tendency to over-predict Grade 0 becomes an advantage, suppressing its primary weakness.

The HiResCAM analysis provides qualitative support for these findings, showing that the CE model consistently focuses on the retinal vascular structure for early grades and on neovascularization regions for severe grades. The CORAL model exhibits more diffuse and less discriminative activation patterns, consistent with its difficulty in distinguishing adjacent severity levels.

Analyzing EfficientNetV2-RW-T models, cross-validation results show that partial fine-tuning significantly outperforms linear probing configurations, with the CE approach achieving a particularly robust improvement in overall metrics. The evaluation of the partial fine-tuning on the test set highlights a clear divergence between the two optimization strategies, showing that the CE framework outperforms the CORAL approach across all global metrics, with superior accuracy (0. 5450 vs. 0. 4850), AUC (0. 8551 vs. 0. 7503), QWK (0. 7076 vs. 0. 6452), and a notably higher Macro F1-score (0. 4777 vs. 0. 2939). This quantitative superiority is strongly supported by the confusion matrices: while CE effectively handles class imbalance by maintaining a stable error distribution and confining misclassifications to adjacent severity stages without dangerous false negatives, CORAL suffers from extreme predictive polarization. CORAL collapses on intermediate grades and heavily biases predictions toward the healthy majority, creating severe clinical safety risks through high false-negative rates for early disease stages.

When shifting to a binary screening paradigm (non-referable vs. referable), both models show strong overall discriminative capacity (AUC of 0.902 for CE versus 0.911 for CORAL). By setting a high sensitivity requirement ($Se \geq 0.95$), the Cross-Entropy model requires a lowered threshold of 0. 202, yielding a sensitivity of 0.972 and an NPV of 0.952, but suffering from lower specificity (0.455) and higher false positives. Conversely, while CORAL targets the same ≥ 0.95 requirement at Thr = 0.243, its sensitivity on the unseen test set drops to 0.906. Although it delivers higher specificity (0.655) and superior precision (PPV = 0.682), this drop reveals that CORAL's inherent polarization makes it less reliable in guaranteeing strict safety thresholds for screening. HiResCAM analysis has shown that Block 2 of the network provides superior, high-resolution localization of clinical retinal pathologies compared to deeper layers. Furthermore, these heatmaps reveal that model errors stem from a complete breakdown and diffusion of feature attention, exposing a critical safety vulnerability for clinical deployment.

The dataset size and diversity represent a clear limitation, with the RetinaMNIST dataset providing only 1,600 fundus images. The small validation set of just 120 samples is particularly limiting for threshold tuning and early stopping, potentially introducing variance in the selected operating points. Furthermore, the dataset is derived from a single source (DeepDRiD), so generalization to other clinical settings or camera systems could be problematic.
Class imbalance also poses inherent challenges for both training objectives. While weighting strategies mitigate this issue to some extent, the CORAL model's collapse on Grade 1 suggests that ordinal approaches are particularly sensitive to imbalance at the extremes of the severity scale.

It is important to highlight that training runs across both models frequently trigger early stopping. Even when stopping criteria are not strictly reached, the loss quickly plateaus and stalls in the final epochs, as both models' high effective capacity rapidly saturates relative to the small training dataset size.

10. Conclusions and Future Work

This project evaluated two training strategies, Cross-Entropy loss and CORAL ordinal regression, for diabetic retinopathy grading on RetinaMNIST using both a lightweight custom CNN (<2M parameters) and a fine-tuned backbone of EfficientNetV2-RW-T (13.6M parameters).
The results confirm a clear trade-off. In both models, the CE approach achieves superior macro-averaged metrics and more balanced per-class performance, with clinically meaningful sensitivity for both early and severe grades. At the same time, CORAL configurations struggle with intermediate classes, particularly Grade 1, which is almost entirely misclassified as healthy. In the binary screening scenario ($Se \geq 0.95$), CORAL outperforms CE in specificity and PPV when applied to the lightweight MicroSENet, suggesting ordinal approaches can be effective for referable disease detection, provided they do not compromise test-set sensitivity. While HiResCAM analysis for the custom CNN indicates that CE focuses more sharply on clinically relevant features compared to the less discriminative patterns of CORAL, the EfficientNetV2 models exhibit a different trend: both the Cross-Entropy and CORAL variants produce remarkably similar attention patterns and are similarly prone to catastrophic errors when feature focus completely breaks down.
Overall, across both architectures, the CE model emerges as the more robust choice for multi-grade classification, also providing interpretable activations that could assist specialists.
Under CE loss, while EfficientNetV2 achieves higher global metrics, MicroSENet demonstrates superior ordinal alignment with a higher QWK. Unlike EfficientNetV2, which exhibits coarser errors on severe cases, MicroSENet provides a safer, more progressive error distribution. Clinically, a high QWK represents strong ordinal agreement that heavily penalizes drastic misclassifications while

tolerating adjacent-grade errors. This ensures mistakes remain safely bounded within neighboring levels, making the custom MicroSENet a more suitable decision-support tool for patient care.

Some immediate directions could extend this work:

- **Hybrid loss function.** Combining CE and CORAL losses into a single objective could leverage the balanced class representation of CE while retaining the ordinal structure enforced by CORAL, potentially mitigating intermediate class collapse.
- **Real-world generalization.** Evaluating these architectures on larger datasets (e.g., EyePACS 2015; APTOS 2019) and out-of-distribution images (e.g., artifacts, unseen cameras) is essential to robustly assess trade-off consistency and verify that the models learn truly generalizable features.
- **Generative augmentation.** Building upon the preliminary exploration by Gangari et al. (2026) on Grade 4 images, extending GAN-based synthetic image generation to all minority classes could further address dataset imbalance and improve overall model robustness.

11. Ethics and Data Privacy

All data used in this project is publicly available and permitted for academic research. RetinaMNIST (part of the MedMNIST v2 collection) is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. Proper attribution to both MedMNIST and DeepDRiD is required. Regarding third-party code, additional packages beyond the default Google Colab environment include torchinfo (MIT), coral-pytorch (MIT), grad-cam (MIT), timm(Apache 2.0), and medmnist (Apache 2.0). All are permissively licensed and allow use with proper attribution.

12. Code and Reproducibility

The code developed for this project is organized into two Jupyter notebooks: scratch.ipynb (custom CNN experiments) and EfficientNet.ipynb (EfficientNetV2-RW-T experiments), both designed to run on Google Colab. Both notebooks share the initial sections on exploratory data analysis, preprocessing, and data augmentation, followed by model-specific training pipelines. The operating point selection and HiResCAM explainability sections are also shared at the end of each notebook. To ensure full reproducibility, every training and evaluation cell can be re-run independently and will produce identical results. This is achieved by fixing random seeds for all stochastic processes: Python's random, NumPy, PyTorch, and DataLoader worker initialization. A dedicated PyTorch Generator is instantiated per training cell to guarantee deterministic data shuffling across runs.

Dependencies (also saved in requirements.txt):

- medmnist (v3.0.2): dataset loading
- torchinfo (v1.8.0): model summary visualization
- coral-pytorch (v1.4.0): CORAL layer and loss
- grad-cam (v1.5.5): HiResCAM explainability
- timm (v1.0.28): EfficientNetV2-RW-T pre-trained weights

All other dependencies (PyTorch, torchvision, scikit-learn, matplotlib, OpenCV, Pillow etc.) are included in the default Google Colab environment.

13. References

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